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COMMENTARY



Urban Resilience Through Cognitive Computing Systems

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ABSTRACT

Global urbanization and heat-related fatalities are rapidly increasing, while the full potential of advanced information technologies to mitigate urban heat has not yet been used by city planners and policymakers. Cognitive computing systems (CCSs), which mimic human information processing and reasoning abilities, can transform how cities strategize and operate. In this commentary, we develop a framework and outline future research directions to understand the disconnect between a strategic and operational CCS, which can facilitate an effective interplay between urban policy (e.g., framing alternative policies) and urban technology (e.g., engaging local communities with data generated by sensor networks).

KEYWORDS

urban resilience; urban policy; urban technology; cognitive computing systems (CCS); sustainability

Urban Heat as a Major Challenge to Making Cities More Resilient

Science tells us that to keep global temperatures from rising by more than 1.5°C, we need to achieve net-zero emissions by mid-century. Sustainable and equitable urban cooling must be a part of cities' efforts to reach net-zero energy targets.—Inger Andersen, Executive Director of the UN Environment Programme (2021)

Rapid global urbanization has increased the vulnerability of cities, which now face more frequent heatwaves, storms, and floods (Selby and Desouza, 2019). The July–August 2003 European heatwave caused 72,210 fatalities from heat-related deaths in major European cities; the economic damage accounted for over US\$12.1 billion (EM-DAT, 2024). More recently, extreme heatwaves have caused wildfires, droughts, and deaths globally, including in Europe, Africa, Asia, and America (NASA, 2022). For the first time in 2022, the UK Meteorological Office issued a red warning for heat, its most extreme alert (Cappucci and Samenow, 2022).

Heat kills more people than any other extreme weather event, because heatwaves are now more intense (Calkins et al., 2016). Urban heat islands (UHI) in metropolitan areas

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can cause city temperatures to be 2–12°C higher than surrounding rural areas (Li et al., 2019). More than 40 million people in the United States live in UHI regions, significantly affecting health and equity (Sharma, 2023). People residing in low-income neighborhoods are particularly stuck in UHI regions compared to their wealthier counterparts (Newsome, 2023). The UHI effect is strongest in densely built-up areas, which absorb heat because vegetation and water bodies have been replaced by impervious materials, such as pavements, concrete buildings, and asphalt roads (Estrada et al., 2017). For cities in hot, warm, or temperate climates, UHI can increase energy consumption, elevate emissions of air pollutants and greenhouse gases, compromise human health and comfort, and impair water quality (EPA, 2023).¹

Natural hazards such as floods and heatwaves often result in substantial immediate destruction of resources and the disruption of services (Desouza and Flanery, 2013). However, climate deterioration is a function of slow-burning factors that seldom get the requisite attention because their daily impact is often indiscernible. Advanced information technologies, which play an increasingly transformative role in city operations (Chin and Guthrie, 2023), can alleviate intergenerational discounting, where investments for urban cooling and climate improvement will not generate benefits for another generation (Jacquet et al., 2013). Society must react to exogenous shocks and make the necessary changes to address threats to resilience proactively. Advanced information systems can play a vital role in enabling urban planners to assess, analyze, simulate, and visualize the impacts of their plans on UHI.

Attempts to address UHI reveal the effects of conflicting priorities between social, environmental, and economic interests. For example, low-income neighborhoods often have higher building densities and fewer vegetative resources, such as parks. This exposes their residents to higher health hazards during periods of extreme heat (Hsu et al., 2021) and energy demand for cooling is usually higher in these areas. Urban resilience requires policies that mitigate UHI by balancing the demand for buildings with more green spaces (Tuczek et al., 2022).² However, the ramifications of such policies can take years or even decades to unfold due to the long-term nature of climate impacts.

Urban planning policies to maximize the desired impacts and minimize the undesirable outcomes can exploit technological advances such as artificial intelligence, predictive modelling, and environmental monitoring (Corbett and Mellouli, 2017). Smart cities are already laden with various technologies to collect and analyze unprecedented data volumes (Brandt et al., 2018). However, current practices fail to harness their full potential. One general class of technologies, cognitive computing systems (CCSs),³ which can mimic human information processing and reasoning capabilities, can transform how cities strategize and operate.

Integrating Cognitive Computing Systems in Urban Design Problems

Growing urbanization increases the complexity of cities, which require advanced information systems to achieve efficient real-time matching and coordination of tasks and resources to enable effective demand and supply responses (Ketter et al., 2020). CCSs can address such needs because they can mimic five fundamental systems typically deployed by organizations to create value (Watson and Pitt, 2022): systems of framing and inquiry for strategic purposes, systems of engagement and production for

Table 1. Types of systems

Type	Purpose	Mechanisms
Engagement	Collaborating and coordinating activities	Touch, gesturing, speech, and writing
Production	Creating products and services	Standard operating procedures to increase efficiency and product consistency
Framing	Justifying the reason for behaving in a particular way or promoting a particular opinion	The use of mental models to focus discussion, justify a viewpoint, and generalize knowledge
Inquiry	Generating knowledge	Use of replicable methods to justify or validate an inference
Record	Recording and retrieving data	Tables, ledgers, and written reports

Source: Watson, 2021

operational procedures, and systems of record at the intersection to enable strategic and operational coordination (See Table 1).

Systems of engagement support collaboration and coordination to achieve shared goals, while systems of production allow humans to create products and services. Systems of framing are used to justify human behavior and decision-making, and systems of inquiry generate knowledge using replicable methods. Systems of record allow storage and retrieval of data (Watson, 2021). While humans have developed these five systems over many years, CCSs now incorporate recent digital technologies, such as cloud computing, the Internet of Things (IoT), machine learning, and big data analytics to save time and effort and reduce human error (Gupta et al., 2018). Early CCSs include IBM's Deep Blue, chess-playing software that defeated world champion Garry Kasparov in 1997 (although it lost the first match in 1996), and Apple's Siri, a virtual assistant introduced in 2011 that uses natural language processing to process voice commands. Today, CCSs are more advanced, as we see with chatbots based on large language models.

A CCS can support an effective interplay between urban policy and technology (See Figure 1), alleviating urban heat and leading to desirable social, environmental, and economic outcomes.

The *Digital Urban Climate Twin (DUCT)*, currently being developed as part of the *Cooling Singapore* project (Ruefenacht and Acero, 2017), is an exemplar CCS. This digital twin allows urban planners to conduct what-if analyses (system of inquiry) to evaluate alternative policies for heat mitigation strategies and justify their decisions (system of framing).⁴ Another example is the *National Integrated Heat Health Information System (NIHHIS)*.⁵ A collaboration with Climate Adaptation Planning and Analytics (CAPA) Strategies has yielded *Heat Watch*, a citizen science project. Residents are equipped with temperature and humidity sensors to attach to their cars and bikes (system of production) for generating fine-granular heat mapping (system of record). Heat maps, created in over 50 US cities, help to identify needs for extreme heat services (system of production). DUCT and NIHHIS require sensor networks to collect and process data for informed decision-making (system of inquiry).

There is often a disconnect between a strategic and operational CCS. However, there is potential to facilitate the interplay between urban policy (e.g., framing alternative policies using digital twins) and urban technology (e.g., engaging local communities to create heat maps). The conjunctive use of urban policy and technology (Degirmenci et al., 2021) afforded by a joint strategic and operational CCS can mitigate urban heat, and, in turn, alleviate social vulnerability, environmental burdens, and economic loss (See Figure 1). There are three areas in our framework: (1) a CCS facilitating the interplay

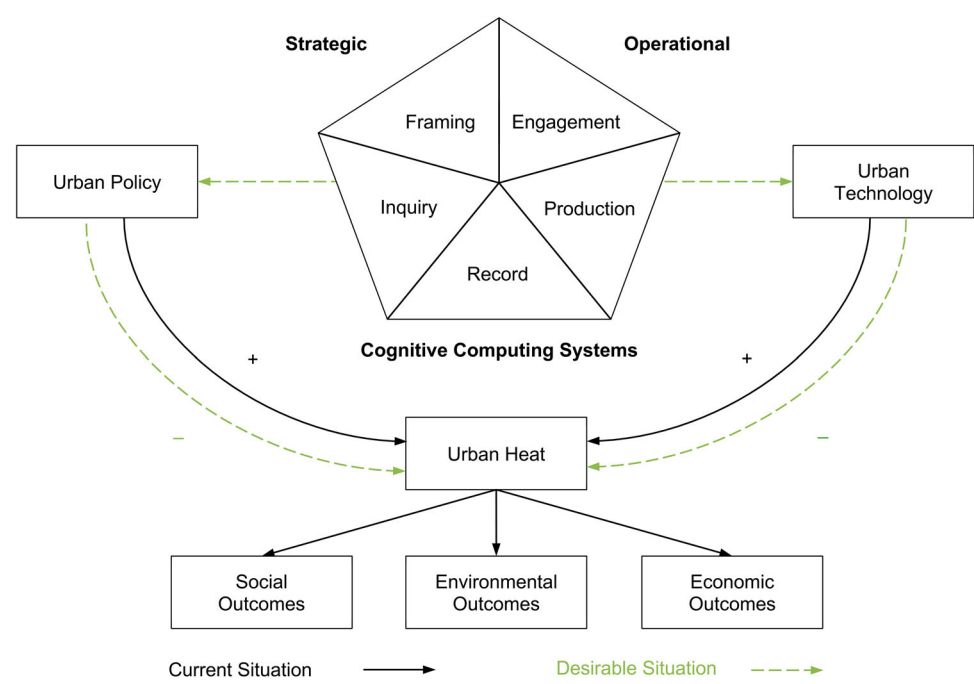


Figure 1. Framework to mitigate UHI through cognitive computing systems

between urban policy and technology; (2) the conjunctive use of urban policy and technology to mitigate urban heat; and (3) outcomes of urban heat as a triple-bottom-line responsibility: social (people), environment (planet), and economic (profit). When computing economic outcomes, it is crucial to recognize that public losses (externalities such as carbon emissions) are partially or fully paid for by public financial means, which are funded by taxes (a private loss) on personal gains.⁶ There can also be a gain from improving the ecological desirability of an area because of cooler temperatures, cleaner air, and environmental attractiveness. The goal is to provide a societal gain to get broad buy-in and lessen the power of vested interests to prevent or delay UHI mitigation.

Future Directions of Cognitive Computing Research for UHI Mitigation

There are various ways to link CCSs directly to the three outcomes. For example, when it comes to *engagement*, CCSs can focus on technologies like edge computing, sensors, and (social) platforms to build connections across layers of society, from individuals to organizations, on UHI issues. With *framing*, a CCS concentrates on simulating the consequences of design choices and their outcomes in the near and long term because decision-makers, the framers of action, understand the costs and benefits. For *production*, a CCS can focus on technologies like robotic process automation to streamline the environmental protocols for new developments. *Inquiry* involves methods like deep learning to discover unknown patterns, which can be linked to digital twins and heat maps to investigate urban fragilities in real-time. Finally, for *records*, sensor networks are required to collect and process data to feed a CCS. To guide future studies on urban resilience through CCSs, we present 15 research questions (See Table 2) covering

Table 2. Future research guide

	Social	Environment	Economic	Study suggestion	Key approach
Engagement	RQ1: What types of CCSs effectively engage community leaders in reducing UHI?	RQ2: How can a CCS nudge residents toward more sustainable behaviors?	RQ3: How can a CCS inform investors about the losses due to UHI?	Implement system dynamics and agent-based simulations to analyze the overall behavior of urban systems and interactions of individual agents	Establish intelligent communication systems that disseminate critical information effectively
Production	RQ4: How do we introduce CCS-driven procedures to alert urban planners to social vulnerabilities?	RQ5: How do we introduce CCS-driven procedures to address urban design problems?	RQ6: How do we introduce CCS-driven procedures to alert urban planners to economic vulnerabilities?	Perform action design research to improve knowledge dissemination through CCS-driven procedures for urban planning	Create digital solutions by integrating a CCS into a cohesive platform fostering collaboration among stakeholders
Framing	RQ7: How can a CCS be used to design policies to justify distributive energy justice?	RQ8: What form of a CCS will help residents identify practical actions for reducing UHI?	RQ9: How do we create a CCS that enables key stakeholders to understand sustainable urban planning possibilities and reduce intergenerational discounting?	Conduct experimental studies to evaluate the effectiveness of CCS implementation for urban policy and technology interventions	Develop policies to enhance CCS implementation into existing and new urban resilience frameworks
Inquiry	RQ10: How can we design a CCS to identify social vulnerabilities?	RQ11: How can UHI modelling be improved with a CCS?	RQ12: What type of CCS would nudge urban planners and investors to lower UHI outcomes?	Undertake data mining studies for predictive modelling based on historical and real-time data	Identify vulnerable areas and analyze UHI trends and consumption patterns
Record	RQ13: What demographic and psychographic data are needed to build a CCS?	RQ14: What data does a CCS require to estimate UHI environmental outcomes?	RQ15: What economic and natural capital data are needed to build a CCS?	Execute big data analytics to provide a structured and organized representation of relevant data for CCS integration	Apply database management, systems design and implementation, and data analytics for integrated CCS functionality and performance

the three triple-bottom-line impact factors and the five CCS areas, study suggestions, and key approach considerations.

Urban studies disciplines can contribute significantly by creating evidence-driven solutions to tackle UHI. Doing so will require collaboration with scholars from domains such as information systems, public policy, economics, and engineering. Citizen science and community engagement are areas that can offer existing frameworks and theories to build solutions around a CCS for *engagement*. Science communication and data visualization are natural allies to inform a CCS for *framing*. The urban discipline could co-create a CCS for *production* and *inquiry* by working with environmental scientists and sustainability advocates. Given the significant expertise within the urban discipline around smart cities, IoT in urban environments, and geographical information systems (GIS), it should lead the conversation for creating standards to promote the interoperability of a CCS focused on *records*.

Notes

1. In cold climates, UHI can have positive effects, such as reduced demand for domestic heating (Roth, 2013).
2. The study proposed an optimization model for Brisbane in Australia and showed that an increase of buildings and decrease of vegetation coverage led to revenue growth by about three billion US dollars based on land valuation data, but the maximum UHI intensity level rose from 4°C to 5°C (Tuczek et al., 2022).
3. Cognitive computing is “a discipline that links together neurobiology, cognitive psychology, and artificial intelligence” (Valiant, 1995: 2).
4. Strategies for mitigating urban heat include increasing vegetation (e.g., trees) and water body coverage (e.g., ponds on roofs); optimizing urban geometry (e.g., varying building forms to improve wind capture); using light-colored surfaces and reflective materials; increasing shading, blinds, and shutters; boosting public transport use and promoting active mobility like cycling; lowering energy consumption by using energy-efficient equipment (Roth, 2013; Ruefenacht and Acero, 2017).
5. <https://www.heat.gov/>
6. The costs of many externalities, such as carbon emissions, will be paid by future generations.

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Disclosure Statement

No potential conflict of interest was reported by the author(s).

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